Spotify Spain Top 50: A Data-Driven Analysis of Song Lifecycle, Artist Dominance, and Chart Movement Prediction

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Abstract

Music streaming charts provide a useful source of information for studying listener preferences, artist visibility, and the changing popularity of songs. This study analyses Spotify Spain Top 50 chart data to investigate artist dominance, song lifecycle patterns, popularity distributions, and the feasibility of predicting subsequent chart positions using historical observations. The dataset covers May 18, 2024, to November 27, 2025. Following data cleaning, the analytical dataset contains 27,750 records across 555 chart dates. Exploratory analysis was conducted to examine artist appearances, song longevity, and chart-position behaviour. Machine learning experiments used chronological train-test splits to reduce the risk of future information leaking into model training. Linear regression and random forest regression were compared against a persistence baseline, which predicts the next position using the current position. In the initial comparison, the persistence baseline achieved a mean absolute error (MAE) of 1.952 positions, outperforming linear regression (2.024) and random forest regression (2.058). In a separate rolling-feature experiment, the baseline again outperformed random forest regression. A movement-classification experiment showed moderate balanced accuracy but limited precision for identifying rising and falling songs. These findings suggest that recent chart position is a strong reference point for short-term prediction, while predicting meaningful chart movement remains challenging. The study also identifies data-quality, temporal-alignment, and reproducibility limitations that should be addressed in future work.

Keywords: Spotify charts, music analytics, song lifecycle, artist dominance, time-series prediction, random forest, chart movement, exploratory data analysis.

1. Introduction

Digital music platforms generate large volumes of behavioural and ranking data. Publicly available chart records can be analysed to understand which artists appear frequently, how songs persist in ranked lists, and how chart positions change over time. Such analysis can support music-industry research, artist strategy, playlist analysis, and the development of predictive analytics tools.

However, chart data presents several analytical challenges. Chart position is relative: a song can move down even if its underlying listener engagement remains strong, simply because other songs perform better. A song's first and last observed chart dates do not necessarily represent its true release or popularity lifespan. Moreover, consecutive chart observations are temporally dependent, making random train-test splitting potentially misleading when the objective is forecasting.

This project studies the Spotify Spain Top 50 chart using data analysis and machine learning. It combines descriptive analysis with predictive experiments to investigate whether historical chart positions can help estimate subsequent positions and classify chart movement.

1.1 Problem statement

Music charts change over time, and it is not always straightforward to distinguish persistent hits, short-lived trends, and songs experiencing meaningful upward or downward movement. This project examines historical chart behaviour and evaluates whether simple machine learning models can predict next-day positions more accurately than a persistence baseline.

1.2 Objectives

The objectives are:

1. Clean and validate the chart dataset for analysis.
2. Examine artist appearances and chart representation.
3. Analyse observed song lifecycles and position changes.
4. Construct historical features for next-day chart-position prediction.
5. Compare regression models with a persistence baseline using chronological evaluation.
6. Evaluate a classifier for rising, stable, and falling chart movements.
7. Document limitations and recommendations for future modelling.

2. Dataset Description

The dataset used in this project is stored as Atlantic_Spain.csv in the project repository. It contains chart observations for songs appearing in the Spotify Spain Top 50.

The original dataset contains 27,800 records and 10 columns. The cleaned dataset contains 27,750 records after handling a date with duplicate blocks of chart entries.

2.1 Dataset attributes

The dataset includes the following fields:

Attribute	Description
date	Date of the chart observation
position	Song's rank in the Top 50 chart
song	Song title
artist	Artist name as recorded
popularity	Available popularity score
duration_ms	Track duration in milliseconds
album_type	Album classification
total_tracks	Number of tracks associated with the album
is_explicit	Indicator of explicit content
album_cover_url	URL associated with the album cover

The cleaned dataset covers 555 chart dates between May 18, 2024, and November 27, 2025. The mean popularity score is approximately 81.12. The observed popularity range is 0–100.

These attributes allow the project to investigate chart representation, song metadata, observed chart longevity, and temporal rank changes. The dataset does not directly measure streams, unique listeners, audience demographics, or causal effects of playlist placement.

2.2 Data preprocessing

The preprocessing stage included inspecting dataset dimensions, checking missing values, examining date-level record counts, and producing a cleaned dataset.

The original dataset had 27,800 records, while the cleaned dataset had 27,750 records. A date containing two blocks of 50 records required duplicate-block handling. The second block was retained for that date. This choice is a preprocessing assumption and should be revisited if the original source can establish which block is authoritative.

The cleaned data contains no missing values according to the project validation checks. However, 106 records have a popularity score of zero. These records were not independently established to represent genuine zero popularity rather than unavailable or differently recorded values; they should therefore be treated cautiously.

Four calendar dates are absent from the cleaned dataset: March 14, 2025; March 25, 2025; July 11, 2025; and August 13, 2025. Consequently, consecutive rows must not automatically be interpreted as observations from consecutive calendar days.

Artist names also require normalisation. For example, the same artist may appear with different capitalisation. Such differences can artificially split artist appearance counts unless names are standardised.

3. Methodology

The study followed a workflow consisting of data validation, exploratory analysis, feature engineering, chronological model evaluation, and interpretation of results.

3.1 Exploratory data analysis

Exploratory analysis was used to examine the distribution of popularity scores, the frequency of artist appearances, and song-level chart histories. Artist appearances were calculated from chart records, while song lifecycle measures were derived from first and last observed chart appearances.

These measures describe the dataset rather than the complete commercial lifecycle of a song. A song may have been released before its first appearance in the dataset, and it may continue to be popular after its final observed chart appearance.

3.2 Song lifecycle analysis

For each song, the first and last observed chart dates were used to calculate the observed chart span. The number of recorded appearances was analysed separately from the elapsed span.

This distinction matters because a long span does not necessarily imply uninterrupted chart presence. A song may disappear from the Top 50 and later return.

Position changes were also used to identify songs that rose, fell, or remained relatively stable in the rankings. These categories describe movement within the chart rather than absolute changes in streams or listener engagement.

3.3 Prediction task

The regression task was defined as predicting the next observed chart position for songs with sufficient historical information. Since lower numerical positions represent better rankings, the objective was to estimate the numerical position at the next observation.

The study compared:

- **Persistence baseline:** predicts that the next position will equal the current position.
- **Linear regression:** estimates a linear relationship between the engineered features and the target position.
- **Random forest regression:** uses an ensemble of decision trees to model potentially nonlinear relationships.

Mean absolute error was used as the primary regression metric:

[
MAE=\frac{1}{n}\sum_{i=1}^{n}|y_i-\hat{y}_i|
]

Here, (y_i) represents the actual next position and (\hat{y}_i) represents the predicted position. Lower MAE indicates smaller average absolute prediction error, measured in chart positions.

3.4 Temporal evaluation

The main evaluation used a chronological split, with training observations ending on August 6, 2025, and test observations beginning on August 7, 2025. This is more appropriate for a forecasting task than randomly distributing temporally related observations between training and test sets.

Two separate regression experiments were conducted. The initial comparison and the rolling-feature comparison used different test samples. Their scores are reported separately and are not treated as a single directly comparable experiment.

3.5 Rolling features

A second experiment incorporated historical position features, including:

- Current position
- Previous observed position
- Position two observations earlier
- Position three observations earlier
- Recent mean position
- Recent position volatility

These features summarise recent chart behaviour and may help represent momentum or instability. However, the historical observations were not perfectly aligned to calendar days in every case. Some

lag features refer to prior available observations rather than exactly two or three calendar days earlier.

3.6 Movement classification

A separate classification task labelled songs according to their change in chart position. Movement was defined as:

$$
\text{Movement}=\text{Current Position}-\text{Next Position}
$$

A positive value indicates that the song moved toward a better numerical rank. A change of at least three positions in the positive direction was labelled **Rising**; a change of at least three positions in the negative direction was labelled **Falling**; all remaining cases were labelled **Stable**.

The resulting labels were imbalanced:

Class	Records	Percentage
Stable	19,402	78.31%
Falling	3,107	12.54%
Rising	2,266	9.15%

Accuracy alone can be misleading when one category dominates. Balanced accuracy, which averages recall across classes, was therefore considered alongside per-class precision and recall.

4. Results and Discussion

4.1 Artist dominance

After normalising artist-name capitalisation, the leading artist appearance counts included:

Artist	Chart appearances
Saiko	1,234
Myke Towers	1,229
Quevedo	1,104

These counts indicate frequent representation in the observed chart dataset. They do not necessarily represent the number of unique songs, total streams, monthly listeners, or overall popularity across all Spotify users.

Artist appearance counts are sensitive to data cleaning and entity normalisation. Standardising artist names is therefore essential before drawing conclusions about comparative chart representation.

4.2 Song lifecycle

“La Bachata” had an observed span of 551 days between its first and last chart appearances, with 538 recorded appearances during that period. It was not present on every intervening day.

This example illustrates the difference between observed chart span and uninterrupted chart persistence. A lifecycle analysis that reports only elapsed time can overstate continuous presence. Future analyses should report both the elapsed span and the number or proportion of chart dates on which each song was observed.

4.3 Regression results

The initial regression experiment produced the following results:

Model	Mean absolute error
Persistence baseline	1.952
Linear regression	2.024
Random forest regression	2.058

In this experiment, the persistence baseline outperformed both regression models. The results suggest that the current chart position was a strong reference point for predicting the next position, and the tested models did not demonstrate an improvement over simply carrying the current position forward.

A separate experiment with rolling historical features produced the following results:

Model	Mean absolute error
Persistence baseline	1.927
Random forest regression	Approximately 2.033

The rolling-feature experiment reached the same broad conclusion: the random forest did not outperform the persistence baseline on the reported test sample.

The two experiments used different test samples and should not be compared as if they were evaluated on identical records. The evidence supports a limited conclusion: under the tested configurations, the additional model complexity did not produce a lower MAE than the baseline.

4.4 Interpretation of regression performance

A persistence baseline is important in forecasting because it establishes whether a more complex model adds predictive value. When chart positions change gradually, using the current position can be a difficult benchmark to beat.

The results do not establish that machine learning is unsuitable for music-chart prediction in general. They show only that the evaluated models and features did not improve upon persistence under the reported experimental setup.

Future models could incorporate better calendar-aligned lag features, explicit entry and exit indicators, song age, recent movement direction, artist-level activity, and additional external signals where their availability and usage rights permit.

4.5 Movement-classification results

The saved movement-classification results reported approximately 0.500 accuracy and 0.504 balanced accuracy for the random forest classifier. The majority-class baseline achieved approximately 0.747 accuracy but only 0.333 balanced accuracy.

The classifier's per-class results were:

Class	Precision	Recall	F1-score
Rising	0.21	0.50	0.30
Stable	0.90	0.50	0.64
Falling	0.22	0.51	0.31

The majority-class baseline's high accuracy is largely explained by the predominance of stable observations. Its low balanced accuracy indicates poor performance when each class is given equal importance.

The random forest's balanced accuracy was higher, but precision for the rising and falling categories remained low. This means that many songs predicted to be rising or falling did not actually belong to those classes. Such false positives limit the classifier's usefulness for confidently identifying songs undergoing meaningful movement.

Accordingly, the classification results should be interpreted as evidence of a difficult and imbalanced prediction problem, rather than proof of a reliable chart-movement forecasting system.

4.6 Reproducibility

A later rerun under a different scikit-learn environment produced a different movement-classification accuracy from the originally saved result. The original saved result reported approximately 0.500 accuracy, whereas the later run reported approximately 0.527; balanced accuracy remained around 0.503.

The discrepancy means the classifier's exact performance is not yet fully reproducible from the reported information. Potential factors include package-version differences, model configuration, random-state settings, or differences in the evaluation data. The precise cause has not been established and should not be asserted without further investigation.

For reproducibility, the project should preserve a complete dependency specification, random seeds, exact feature columns, training and test date boundaries, and the commands used to generate the results. Saved outputs should be regenerated in a controlled environment before making definitive performance claims.

5. Interactive Dashboard

The project includes a Streamlit dashboard intended to make the chart dataset easier to explore. Its features include date and artist filters, summary metrics, a chart-position visualisation, artist appearance counts, a data preview, and CSV download functionality.

The dashboard should be described as an **interactive analytics dashboard**. The deployed application does not itself execute the trained machine learning models or provide live predictive inference. Therefore, the presence of machine learning experiments in this paper should not be interpreted as a claim that the dashboard offers deployed prediction functionality.

The public dashboard URL should be added to the final manuscript and internship submission once verified.

6. Limitations

This study has several limitations.

First, the analysis uses a single country's Top 50 chart over a limited period. The findings may not generalise to other countries, genres, platforms, or longer time horizons.

Second, chart rank is a relative measure. It does not directly reveal stream counts, listener engagement, revenue, or the reasons behind a song's movement.

Third, missing calendar dates and the treatment of duplicate records may affect temporal analysis. The decision to retain one of the duplicate date blocks should be validated against the original source.

Fourth, artist names and song titles can contain spelling, capitalisation, collaboration, and metadata inconsistencies. These can affect grouping and lifecycle statistics.

Fifth, the lag features do not always correspond to exact calendar-day intervals. A more rigorous forecasting pipeline should explicitly join observations by date and song, account for missing dates, and define how chart entries and exits are handled.

Sixth, the regression models did not outperform the persistence baseline in the reported experiments. The classification model showed weak precision for rising and falling songs.

Finally, the classifier's reproduced performance differed across software environments. A pinned environment and a complete experiment log are necessary before the reported metrics can be considered fully reproducible.

7. Future Work

Several improvements could strengthen this project.

1. **Improve data validation:** verify the source of duplicate blocks, investigate zero popularity values, and explicitly account for missing chart dates.
2. **Improve entity resolution:** standardise artist and song identifiers to reduce duplicate representations.
3. **Refine lifecycle metrics:** report continuous chart streaks, total observed days, interruptions, and re-entry frequency separately.
4. **Improve temporal features:** construct date-aligned lags and rolling statistics using only information available before each prediction time.
5. **Establish stronger baselines:** compare persistence with recent-average, seasonal, and movement-based baselines where appropriate.
6. **Improve movement classification:** evaluate class weighting, calibrated probabilities, and decision thresholds while retaining chronological validation.
7. **Expand the data:** where permitted, include additional chart periods or regions and evaluate whether the findings generalise.

8. **Strengthen reproducibility:** pin dependency versions, record random seeds, preserve experiment configurations, and rerun all reported metrics from a clean environment.
9. **Extend the dashboard:** add clearly labelled historical movement summaries and, only after validation, a separate interface for model predictions.

These steps could improve both the reliability of the analysis and its practical value.

8. Conclusion

This study analysed Spotify Spain Top 50 chart data to investigate artist representation, observed song lifecycles, and the feasibility of predicting chart positions and movement.

The cleaned dataset contained 27,750 records across 555 chart dates. Artist appearance analysis identified Saiko, Myke Towers, and Quevedo among the most frequently represented artists after normalising name capitalisation. Song lifecycle analysis demonstrated why elapsed time between first and last chart appearances must be distinguished from continuous chart presence.

In regression experiments, the persistence baseline outperformed linear regression and random forest regression. A separate rolling-feature experiment reached the same general conclusion. The movement classifier improved balanced accuracy relative to the majority-class baseline, but its precision for rising and falling songs remained limited. A discrepancy between saved and rerun classifier metrics also highlighted the importance of reproducible environments.

The central finding is that chart analysis can provide useful descriptive insights, but reliable short-term prediction requires careful temporal feature construction, robust baselines, and reproducible evaluation. The current results do not justify claiming that the tested machine learning models outperform a simple persistence strategy. Instead, they provide a foundation for further work on better-aligned historical features, data quality, and more reliable chart-movement modelling.

References

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Data and Code Availability

The analysis code and project files are available in the author's public repository:

https://github.com/mirzsreek/UnifiedMentorInter_project

The dataset's original source, licensing terms, and redistribution permissions should be documented in the repository before redistribution. The dashboard link should be added after verifying that it is publicly accessible.

Declaration

This manuscript describes an internship project and has not been represented as a peer-reviewed or published research article. Results and conclusions are limited to the experiments and data described above.